

Pathology of skin tumors

Principles and Practices of Osteopathic Medicine I

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Session objectives

1. Describe and discuss the pathology and morphology (gross and microscopic) of benign epithelial tumors which include **seborrheic keratoses, acanthosis nigricans, fibroepithelial polyps, and epithelial cysts**
2. Describe and discuss the pathology and morphology (gross and microscopic) of the premalignant epithelial lesion **actinic keratosis**
3. Describe and discuss the pathology and morphology (gross and microscopic) of the malignant epithelial lesions **squamous cell carcinoma and basal cell carcinoma**
4. Describe and discuss the pathology and morphology (gross and microscopic) of melanocytic lesions such as **lentigo and various melanocytic nevi including dysplastic nevi**
5. Describe and discuss the pathology and morphology (gross and microscopic) of **malignant melanoma**

Epithelial tumors

- **Non-melanocytic**

- **Benign**

- Seborrheic Keratoses
 - Acanthosis nigricans
 - Fibroepithelial polyp
 - Epithelial cyst

- **Premalignant**

- Actinic Keratosis

- **Malignant**

- Squamous cell carcinoma
 - Basal cell carcinoma

- **Melanocytic**

- Solar Lentigo
 - Melanocytic Nevi
 - Dysplastic Nevi
 - Melanoma

Seborrheic keratosis (SK)

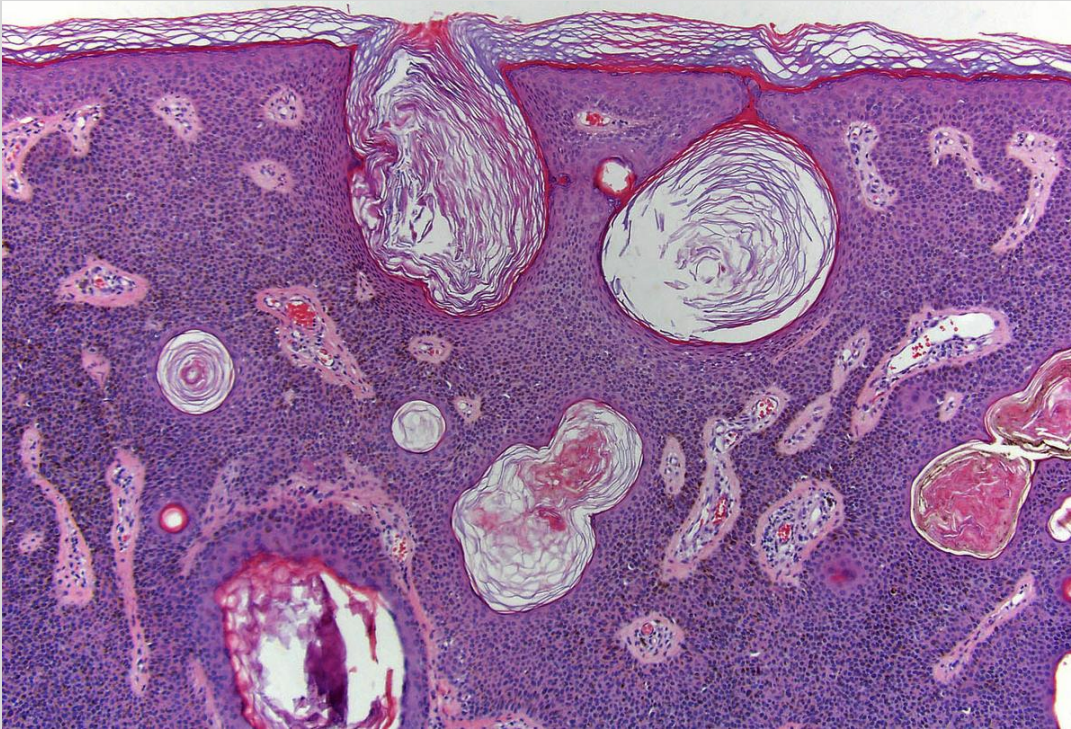
- Common, arise spontaneously
- Occur most frequently in **middle-aged or older** individuals
- Clinically appear as **well demarcated**
 - Round, flat plaques that vary in size
 - Uniform **tan to brown** color
 - Velvety surface
 - Appear “**stuck on**” - can be peeled off
- **Leser-Trelat sign** – Paraneoplastic syndrome w/ rapid increase in number of SKs, marker for adenocarcinomas (Gastric, Colon, Pancreatic)
 - Activating mutations in **FGFR3** (receptor tyrosine kinase) found in many SKs thought to play a role in development



SK histopathology

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- Sheets of benign **basaloid cells**
- Exuberant keratin production (**hyperkeratosis**) at the surface
- Small keratin-filled cysts (**horn cysts**)

Acanthosis nigricans: Overview

- **Thickened, hyperpigmented** regions of skin
- Hallmark regions are **flexural** areas – axillae, skin folds of neck, groin
- ↑ **growth factor** (FGFR3, IGFR-1, EGFR) thought to cause development
- Can be an important cutaneous **sign** of **underlying benign & malignant** conditions
 - Benign (80%) - develops gradually
 - Usually during childhood, puberty
 - Associated with **diabetes** (insulin resistance) & **obesity**
 - Malignant (20%)
 - Usually in middle-aged and older individuals
 - Paraneoplastic phenomenon
 - Underlying **adenocarcinoma** (often GI)

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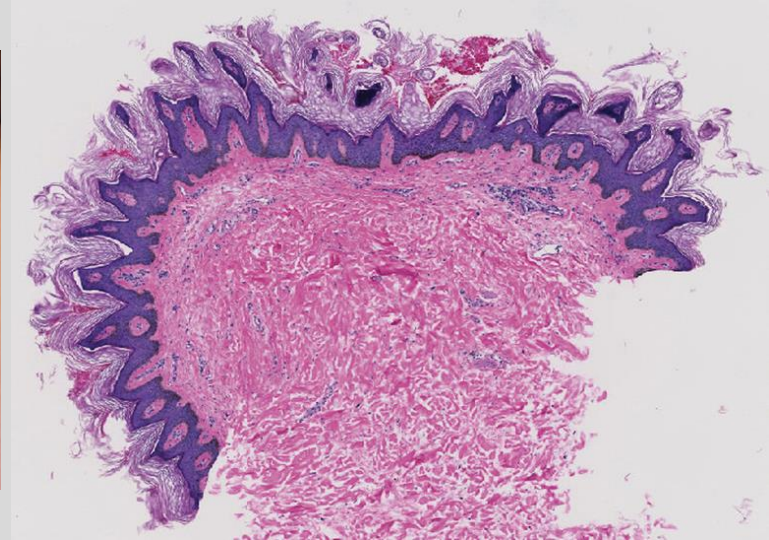
Acanthosis nigricans: Clinical findings and histopathology

- Epidermis and underlying enlarged dermal papillae **undulate** sharply w/ repeating peaks and valleys
- **Hyperkeratosis**
- Basal cell layer **hyperpigmentation** (no melanocyte hyperplasia)



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Fibroepithelial polyp (acrochordon, squamous papilloma, skin tag): Overview

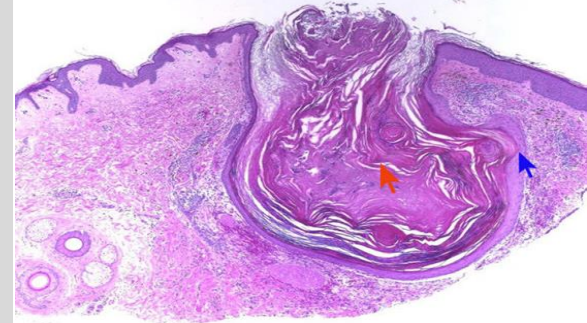
- Very common cutaneous lesion
- **Middle to older age** patients
- Often appear as groups on the neck, trunk, face, intertriginous areas
- Attached by a small, thin **stalk**
- Occasionally associated with **diabetes, obesity, and intestinal polyposis**
- May become more numerous during **pregnancy** due to hormonal stimulation
- Histopathology shows **fibrovascular core** covered by **benign squamous epithelium**



Epithelial/follicular inclusion cyst (Wen)

- Common lesions due to **invagination** and **cystic expansion** of **epidermis** or, more commonly, a **hair follicle**
- Clinically
 - **Well circumscribed**, **firm**, often **moveable**
 - Typically dome shaped, **flesh colored**
- Histopathology shows
 - **Keratin** filled cysts
 - Subtypes
 - Epithelial Inclusion: **wall resembles normal epidermis**
 - Pilar/Trichilemmal: wall resembles follicular epithelium (pilar = **hair**)
 - Dermoid: similar to epidermal cyst, have **multiple appendages**
 - Steatocystoma: wall resembling **sebaceous gland duct**

<https://www.penndermspecialists.com/identifying-treating-cysts-with-surgical-dermatology/>



➡ **Lamellated keratin**

➡ **Squamous epithelium with a granular layer**
(resembles infundibular portion of hair follicles)



[https://link.springer.com/
chapter/10.1007/978-3-
031-64102-2_2](https://link.springer.com/chapter/10.1007/978-3-031-64102-2_2)



<https://contourderm.com/actinic-keratosis/>

Actinic keratosis (AK): Overview

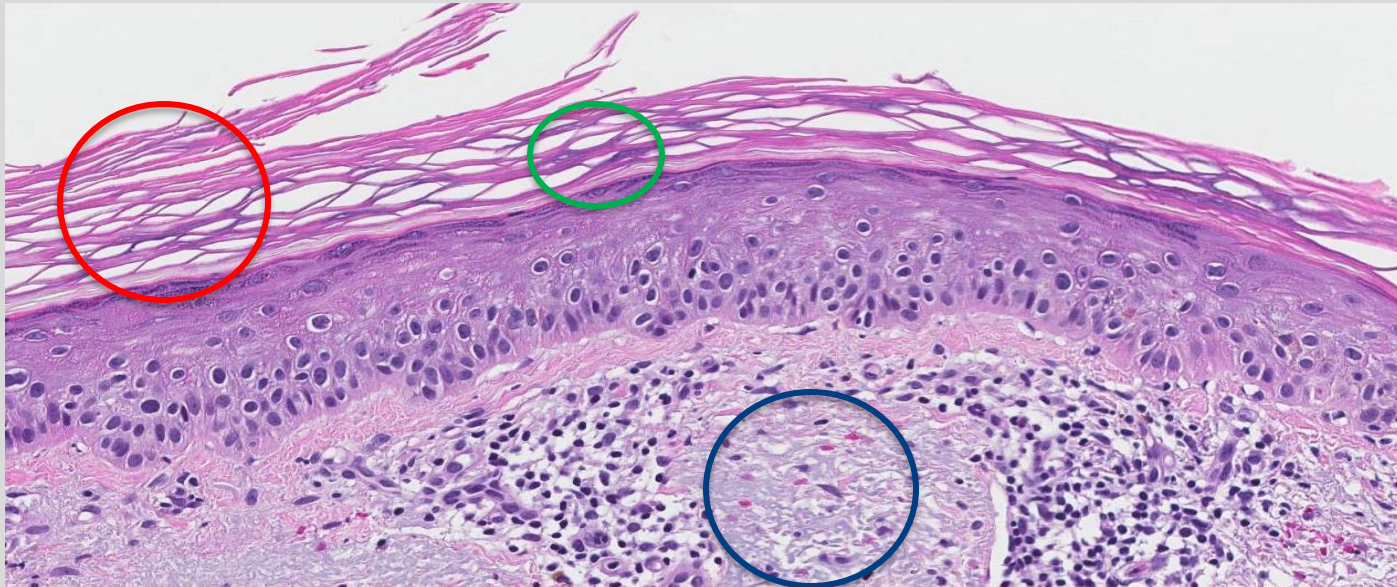
- **Premalignant** epithelial tumor
- Squamous **dysplasia**
- Caused by chronic exposure to **sunlight** (sun exposed areas with sun damaged skin)
- Often appear as **multiple** lesions
- Most often in **lightly pigmented patients**
- Some lesions produce so much keratin that a “**cutaneous horn**” develops (top image)
 - May become so prominent that they resemble actual horns of animals
- Clinical findings:
 - **Small**, <1 cm diameter
 - Can be flesh colored, red or tan-brown
 - Rough consistency (**sandpaper**)

AK: Histopathology

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- Cell atypia – basal layer
- **Hyperkeratosis** – thickening of stratum **corneum**
- **Parakeratosis** – **retained nuclei** in thickened stratum corneum
- **Solar elastosis** – Due to long term **UV** exposure, causing thickened, yellowed and **wrinkled** skin due to **damaged collagen & elastin fibers** with **grey-blue** thickening within **superficial dermis** - abnormal ECM produced by sun-damaged fibroblasts



AK: Precursor lesion

- AKs are well-known precursor to squamous cell carcinoma (**SCC**)
- Can develop over **months to years**
 - Can go away on their own or persist for months to years
 - 1-10% of AKs themselves develop into SCCs
- Progression to **full-thickness** cellular/nuclear **atypia** heralds progression into **SCC in-situ**
- Treatment
 - Gentle curettage and/or freezing
 - Topical chemotherapeutics (5-Fluorouracil, Imiquimod)

Squamous Cell Carcinoma (SCC): Overview

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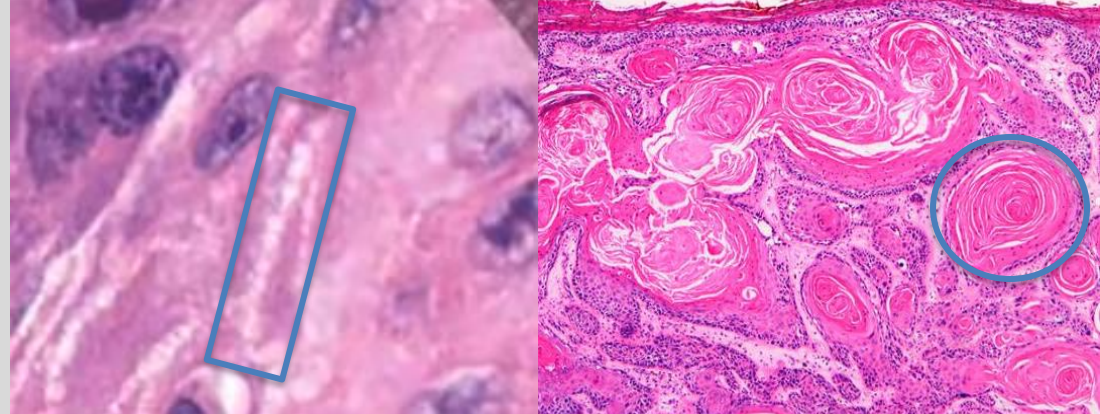
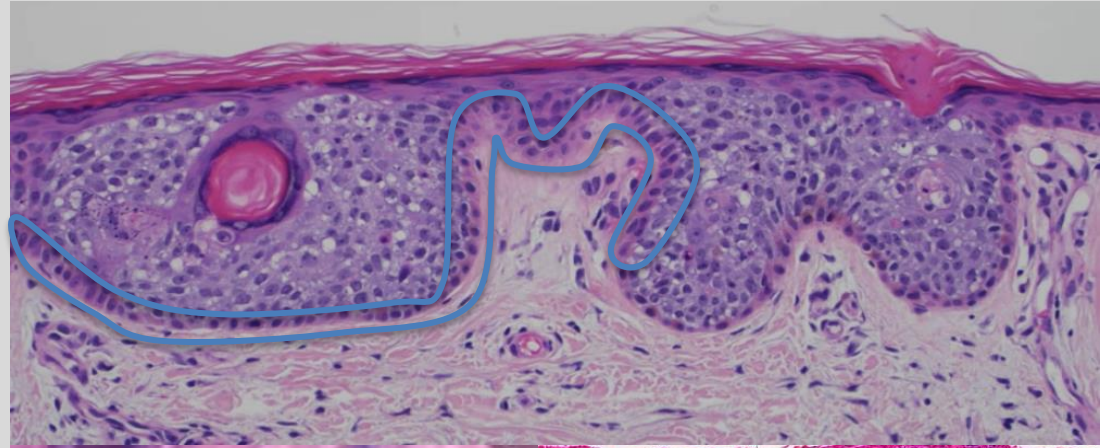
- 2nd most common cancer in the world (>1.5 million cases in US annually)
- **Older** people most affected, **M>W**
- Most important cause is DNA damage from **UV** exposure
 - Roughly proportional incidence to lifetime sun exposure
 - Other risk factors: Industrial carcinogens (tars and oils), chronic ulcers & draining osteomyelitis, old burn scars, arsenical ingestion, ionizing radiation, and (oral cavity) tobacco
- People on **immunosuppression** at an increased risk
- Low rates of invasion and metastases, good prognosis

SCC: Histopathology

- Pathogenesis
 - Incidence of **p53** dysfunction in **AK** found in Caucasians is high, suggesting this is an early event
 - UV light can damage DNA and cause **immunosuppression** to Langerhans cells
- Clinical findings and histopathology
 - Invasive lesions: Marked hyperkeratosis, **keratin pearls**, **intercellular bridging**, nodules and ulceration

SCC: Histopathology

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https://www.researchgate.net/figure/b-Squamous-cell-carcinoma-a-common-skin-cancer_326487921

https://commons.wikimedia.org/wiki/File:Micrograph_of_well-differentiated_and_invasive_squamous-cell_carcinoma.jpg

Basal Cell Carcinoma (BCC): Overview

- **Locally aggressive** cutaneous tumor w/ mutations that **activate Hedgehog signaling pathway**
- **Most common invasive cancer** in humans, very **slow growing (rarely metastasize)**
- **Most common cancer** in humans (~ 2+ million new cases per year)
- Sun (**UV**) exposed areas most effected
 - Most commonly affect **lightly pigment** patients (blonde hair, blue eyes, pale skin)
 - Risk ↑ w/ immunosuppression
- Clinical appearance
 - **Pearly (waxy) papules**, often with **telangiectasias**
 - **Rolled borders**
 - Central ulceration may occur, less common than SCC
- Histopathology
 - Tumor cells **resemble basal layer** of epidermis
 - Form **nodular** lesions (nests) that **infiltrate** into **dermis**
 - Peripheral **palisading**

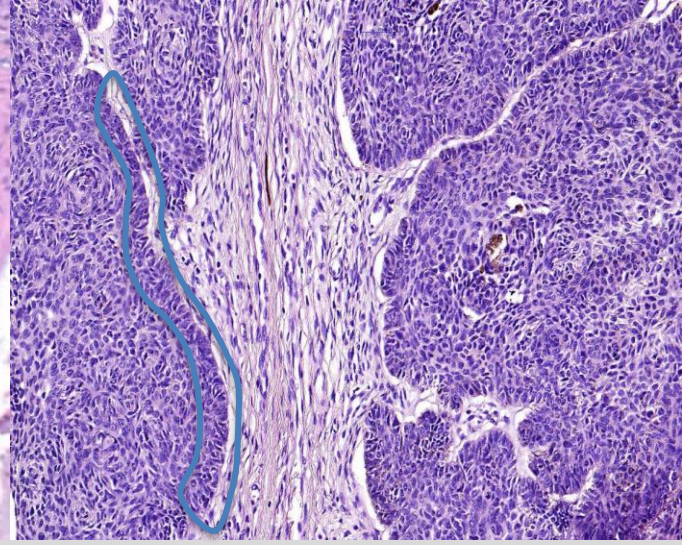
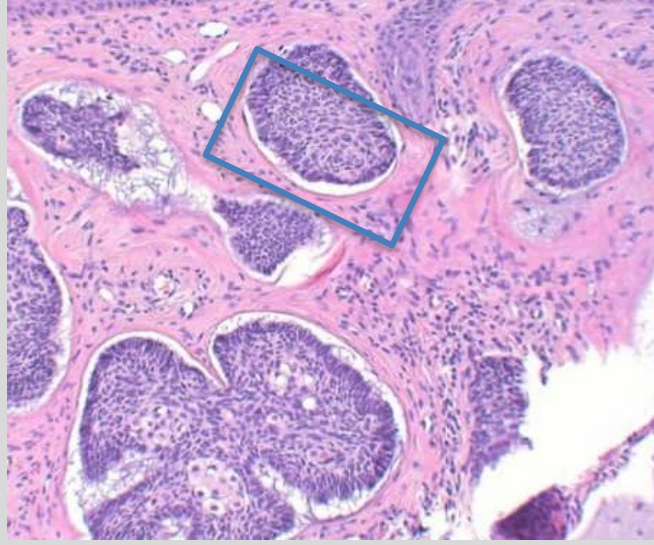
BCC: Pathogenesis and treatment

- Pathogenesis
 - Associated with **mutations that activate Hedgehog pathway**; loss of function of **PTCH1 (receptor for sonic hedgehog, SHH)**
 - Loss of function mutation in PTCH1 → Inability to repress transmembrane protein called SMO → ↑ transcription factor, GLI1 → tumor growth
 - **Gorlin Syndrome**: PTCH1 mutation → ↑incidence of BCCs
- Treatment
 - Surgical Excision
 - Wide Local Excision
 - Mohs Micrographic Surgery
 - Cryotherapy
 - Topicals – Imiquimod, 5-Fluorouracil

BCC: Histopathology

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<https://www.cram.com/flashcards/dermatology-pictures-gross-and-histology-4982763>

<https://dermnetnz.org/image/detail/6705-basal-cell-carcinoma-pathology>

Melanocytic Tumors

- Solar Lentigo
- Melanocytic Nevi
- Dysplastic Nevi
- Melanoma

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Solar lentigo: Overview

- Common **benign localized hyperplasia** of melanocytes
 - Not precancerous, generally do not progress to malignancy
- Exclusively in **sun** exposed areas
- Typically age >40
 - Prevalence ↑ in lightly pigmented individuals
- Typically small (5–10 mm), oval-shaped, tan to brown macules
- Can occur on **skin or mucosa** membranes
- **Lentiginous growth pattern** seen in a subtype of melanomas (lentigo maligna melanoma)
 - Melanocytes proliferate linearly along dermoepidermal junction
 - Rather than forming clusters or nests

Solar lentigo clinical findings & histopathology

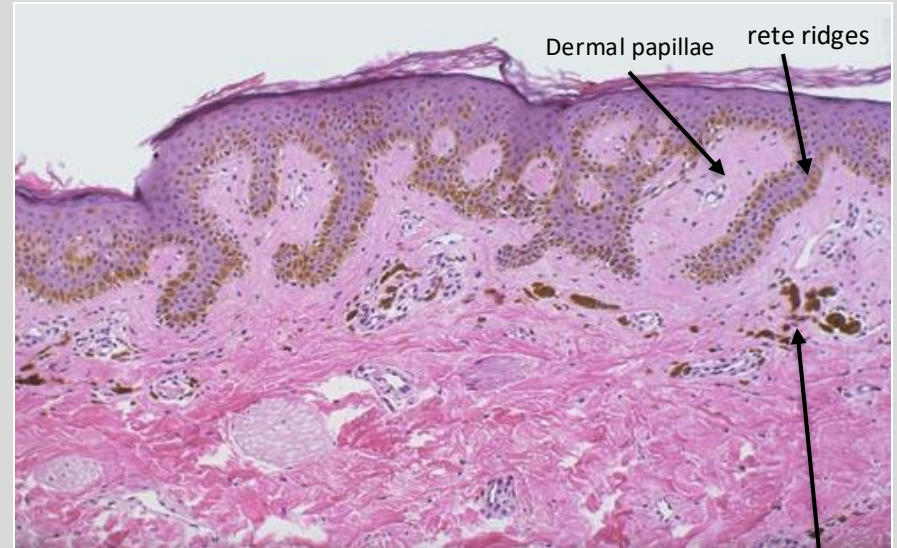
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- **Linear** melanocytic hyperplasia restricted to immediately above the **basement membrane**
- Elongation & thinning of rete ridges



<https://medic.studio/patologicheskaya-anatomiya/koja-67046.html>



<https://basicmedicalkey.com/the-skin-13/>

Melanophages

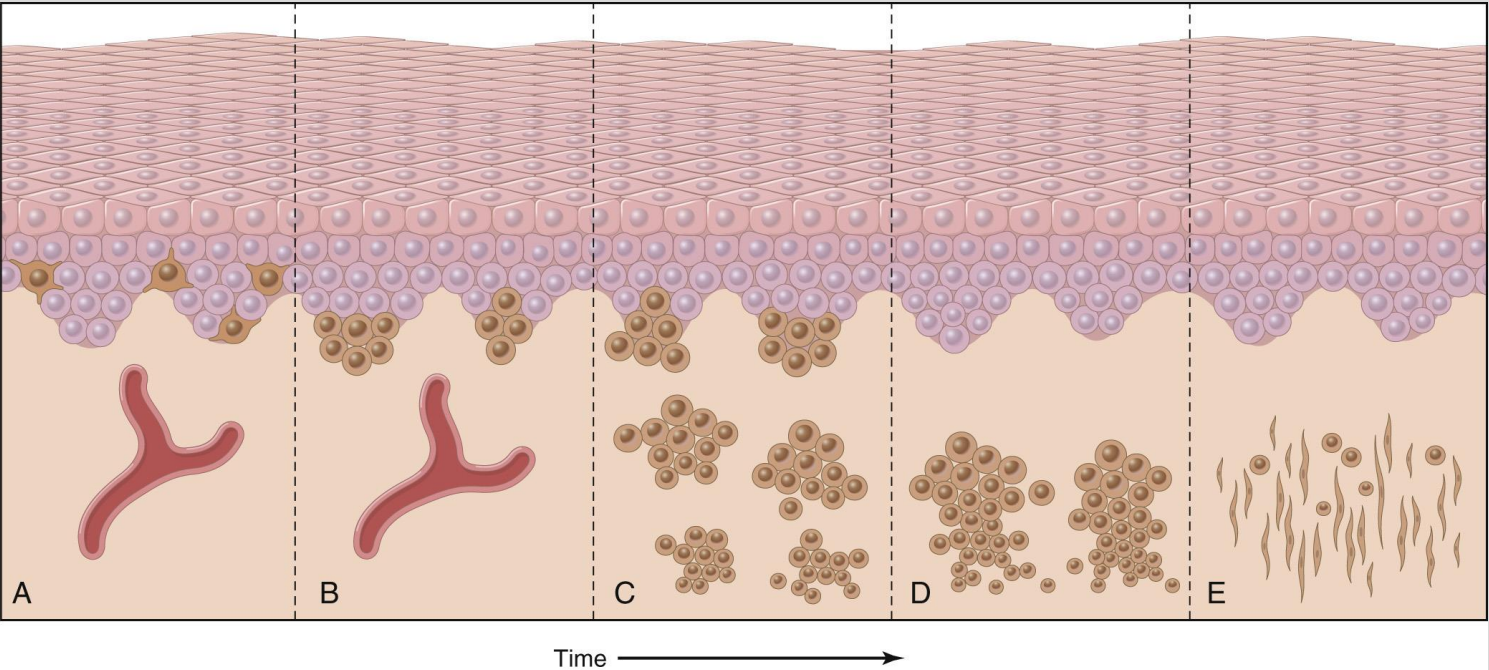
Melanocytic nevus (moles)

- Any neoplasm of melanocytes
- Can be congenital or acquired
- Wide ranging appearance and histology that can change over time
- Typically smaller than malignant melanomas (<6 mm diameter)
 - **Symmetric, uniform** in color, with **smooth borders**
- ~70% have either **BRAF** or **NRAS mutation**

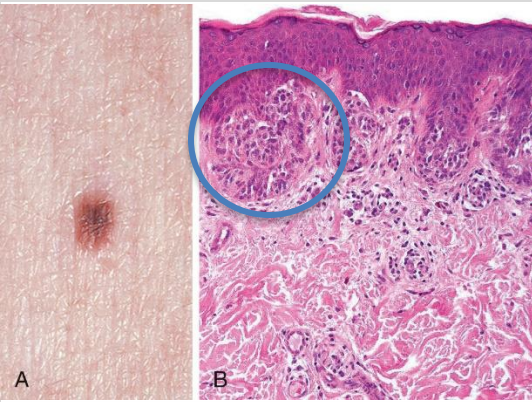
Representative variant forms of melanocytic nevi

Name		Description
Congenital		Present at birth. Can have tuft of dark hair emerging. Can be flat (smaller) or raised (larger). Melanoma can rarely develop within a congenital nevus.
Blue nevus		Acquired, firm dark-blue to grey-black nodule with sharp borders. Localized proliferation of dermal melanocytes .
Spitz nevus		Common in children . May appear similar to hemangioma . Small, dome-shaped, hairless. Often noted for recent, rapid growth.
Halo nevus		Marked by " halo " of depigmentation (leukoderma), surrounding a nevus. Autoimmune response against surrounding melanocytes at the dermal-epidermal junction. Can occur in vitiligo and indicate incipient vitiligo.
Dysplastic nevus		Atypical nevi, potential indicator of future melanoma. Represented by cytological atypia, intraepidermal nests.

Maturation sequence of nondysplastic melanocytic nevi



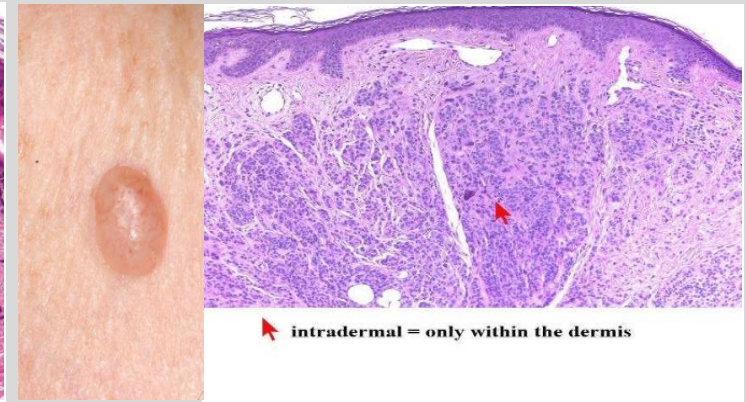
Melanocytic nevi histology



Junctional Nevus



Compound Nevus



Intradermal Nevus

Dysplastic nevus (atypical mole): Overview

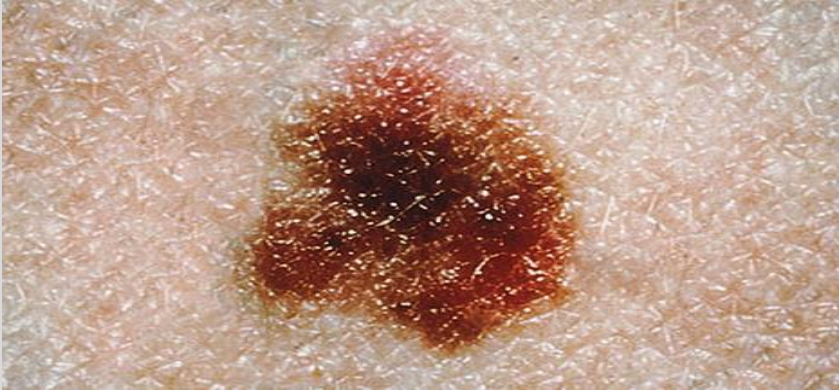
- **Potential precursors** to melanoma
 - Most never progress
 - ↑ risk for melanoma w/ ↑ dysplastic nevi
- Disordered proliferation of **atypical** melanocytes
- Most occur on **exposed skin** (but also covered skin of lightly pigmented individuals)
- Often carry activating **BRAF** or **NRAS** mutations (proto-oncogenes) and loss of function mutations in **CDKN2A** (tumor-suppressor)
- Can be autosomal dominant familial **multiple dysplastic nevi and melanoma syndrome**
 - Hundreds of dysplastic nevi

Dysplastic nevi: Clinical appearance

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- Typically **larger** than other acquired nevi (> 5 mm)
- Can be **flat or raised**, often cracked or pebbly surface texture
- **Variable pigmentation**
- Can have **irregular borders**



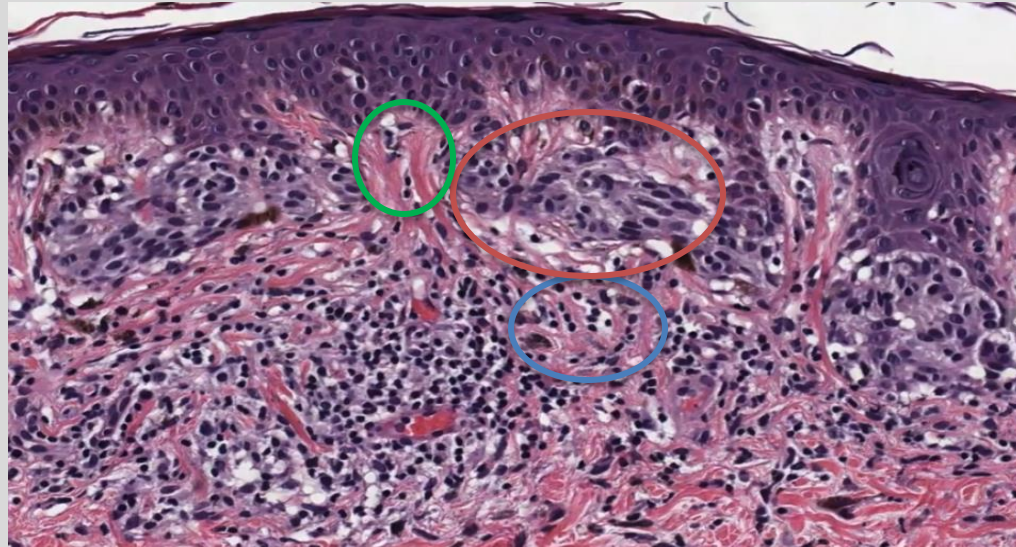
<https://skin-analytics.com/news/skin-cancer/how-to-tell-a-melanoma-from-a-scary-mole/>

Dysplastic nevi: Histopathology

- **Cellular Atypia** – not seen w/ acquired/congenital nevi
 - Irregular shaped cells, enlarged nuclei, hyperchromasia
- **Linear fibrosis** around rete ridges
- Bridging: Adjacent nests of **melanocytes fuse** across tips of **rete ridges**
 - A hallmark of **architectural disorder** (one of the criteria to diagnose a dysplastic nevus)
- **Melanin incontinence**
- **Lymphocytic infiltrates**

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- **Fibrosis**
- **Bridging**
- **Lymphocytic infiltrate**

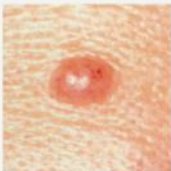
Melanoma: Overview

- Highest mortality of all skin cancers
- Linked to specific mutations (**BRAF, NRAS, KIT, NF1, CDKN2A**) caused by **UV** radiation
- Most are cured surgically
 - Survival drops to ~35% w/ distant **metastases**
 - Early detection is crucial
- Typically asymptomatic
- **ABCDEs** of melanoma clinical findings
 - A – Asymmetrical
 - B – Borders: Irregular
 - C – Color: Variegated
 - D – Diameter: larger than pencil eraser, >6mm
 - E – Evolution: change over time, often rapid

Benign mole

Melanoma

A



Asymmetry

Is the spot asymmetric?
If you draw a line in the middle,
one half does not match the other.

B



Border

Are the borders uneven?
The borders are not well defined
or are irregular in shape.

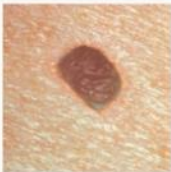
C



Color

Does it have different colors?
Melanoma lesions are often more
than one color or shade.

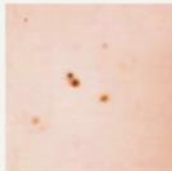
D



Diameter

Is it larger the 1/4"?
Melanoma growths are normally
larger than 1/4" in diameter.

E



Evolving

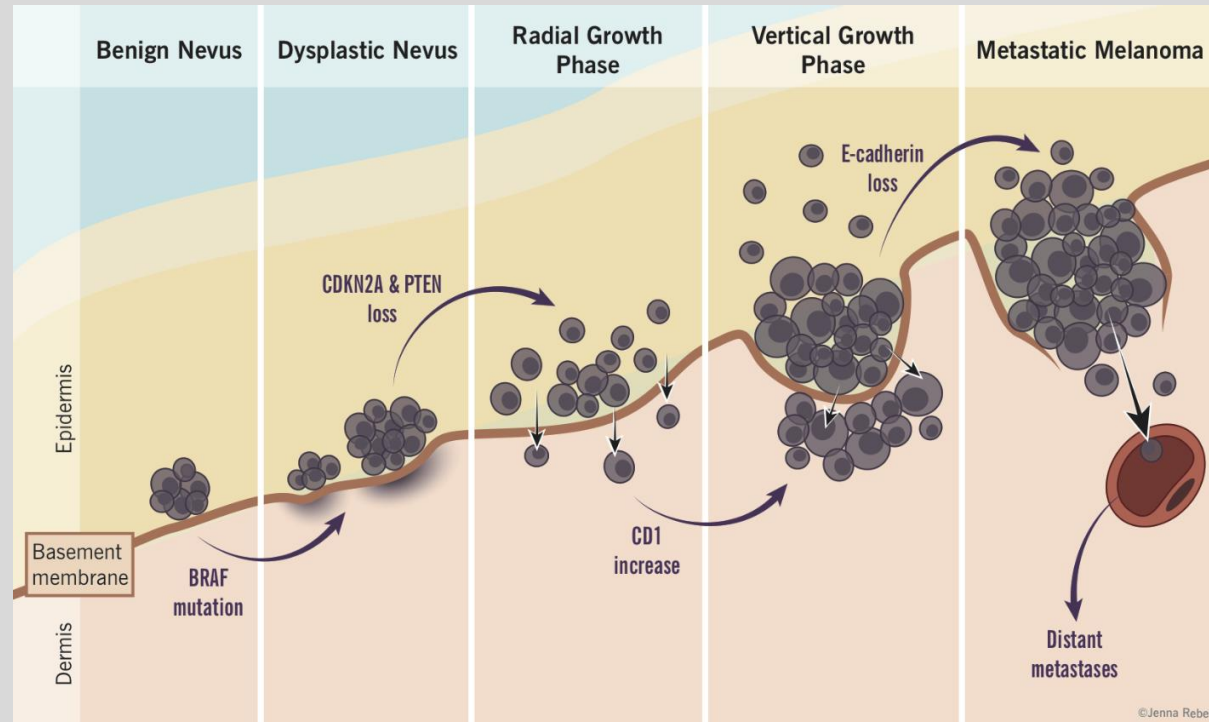
Is there a change in growth?
Over time, Melanoma will often
change characteristics, such as size,
shape or color.

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Melanoma development

- Two distinct periods of growth:
- **Radial growth**
 - Grows along dermoepidermal junction
 - A few cells may invade papillary dermis
 - Can remain here months to years
- **Vertical growth**
 - Invades into dermis
 - Correlate w/ raised areas of lesions
 - Proximity to lymph & blood vessels
 - Avenues for metastasis
 - Poor prognosis



Melanoma major subtypes

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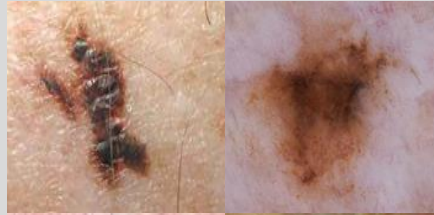
- **Superficial spreading melanoma**
 - Most common
 - **Best** prognosis, long **radial growth** phase
- **Lentigo maligna melanoma**
 - Higher rate in older individuals
 - **Good** prognosis, **radial growth** phase for years
- **Nodular melanoma**
 - Rapid radial growth & **early vertical** growth
 - **Poorer** prognosis, especially if metastasizes
- **Acral lentiginous melanoma**
 - Most common in **non-white patients**
 - Palms, soles, subungual sites most effected
 - Early vertical growth
 - **Poor** prognosis due to **delayed recognition**

Melanoma subtype appearance

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Superficial spreading melanoma



Lentigo maligna melanoma



Nodular melanoma

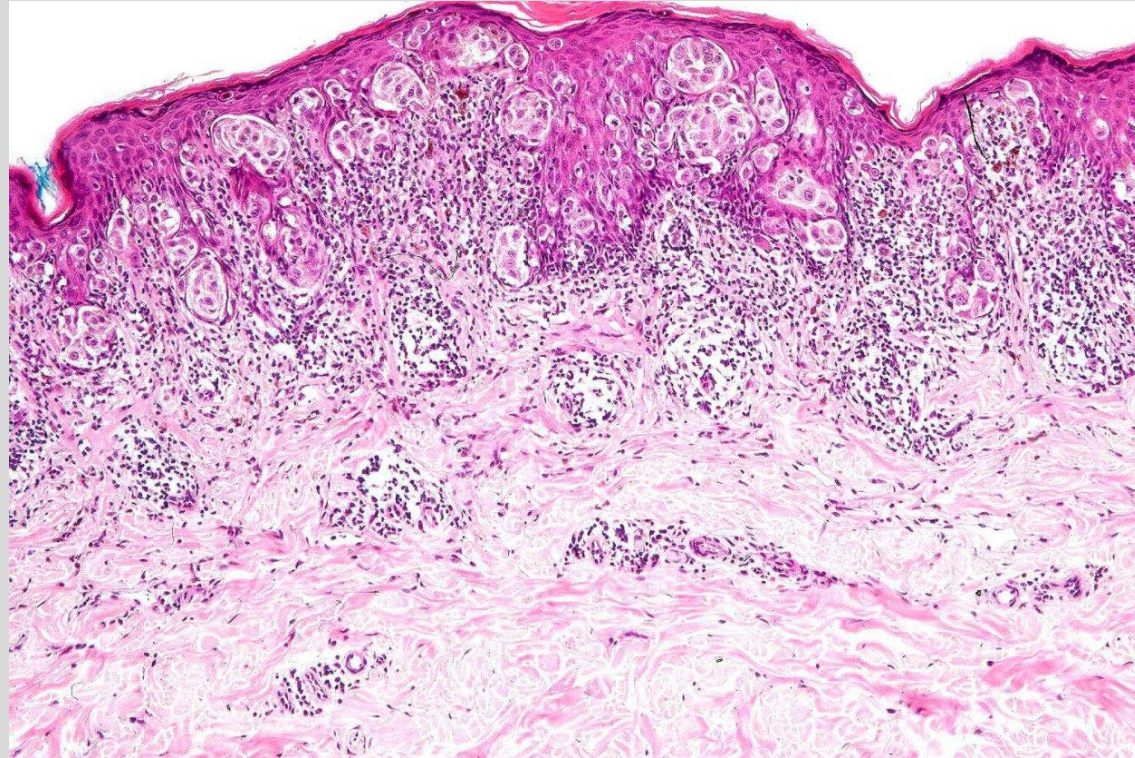


Acral lentiginous melanoma



Melanoma histopathology

- Some key clinical features include:
 - Asymmetry
 - Ulceration
 - Cytologic atypia
 - Pagetoid spread
 - Dermal mitoses



Melanoma diagnosis & prognostic factors

- **Diagnosis**
 - Excisional **biopsy** to confirm diagnosis & determine thickness (**Breslow depth**) and **ulceration**
 - **Sentinel lymph node** biopsy often done for larger tumors to check for local spread
 - IHC staining & genetic testing
- **Prognosis**
 - **Breslow Depth** (measure from epidermal granular layer top surface to point of maximum tumor thickness)
 - Local & distant **metastases**
 - Mitotic rate
 - Ulceration
 - Tumor infiltrating lymphocytes

TNM cancer staging system for melanoma

- **Breslow depth** (thickness) and **ulceration**
 - T1: ≤ 1 mm
 - T2: 1.01 to 2 mm
 - T3: 2.01 to 4 mm
 - T4: > 4 mm (Subcategories: a (no ulceration), b (with ulceration): Ex. 4a – without ulceration, 4b – with ulceration)
- Staging
 - Stage 0 – **in-situ** (confined to epidermis)
 - Stage I-II: **Localized or locally advanced** (thicker - $\uparrow$ risk than Stage I) primary tumor, **no nodal spread or metastases** (T1-T4, N0, M0) $\sim 98\%$ 5-year survival
 - Stage III: Regional lymph **node** involvement (T1-T4, **N1-N3**, M0) $\sim 35\%$ 5 year survival
 - Stage IV: Distant **metastases** (Any T, Any N, **M1**)

Melanoma treatments

- **Excision** of lesion and surrounding normal tissue (wide local excision)
 - **Margins** of tissue evaluated for tumor (desire clear/clean/negative margins)
 - Indicates surgeon likely removed all tumor, reducing risk of cancer recurrence there
 - Often **sentinel lymph nodes** evaluation guides tumor staging
- **Checkpoint inhibitors**: Blockade of inhibitory receptors, CTLA-4 and PD-1, **enhances T-cell-mediated antitumor** immune responses, leading to improved survival
 - CTLA-4 Inhibitors: **Ipilimumab**
 - PD-1 Inhibitors: **Nivolumab** and **Pembrolizumab**
- Targeted therapies in cancers with known genetic mutations
 - BRAF Inhibitors: In patients with BRAF V600E mutations that drives melanoma growth

Lecture and Lab Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>